PROFILE HANDBOOK

Programs A-I

The cartridge-free field manual for Power Glove Vision profiles.

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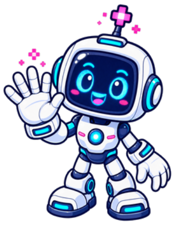
The **PowerGlove Vision Controller (Arduino UNO Q)** recognizes these shared gestures and sends the selected Program's gamepad mapping to RetroPie.

Bad Street Brawler supplied nine programs that configured the original Power Glove for different games. This guide explains how those programs worked and how PowerGlove Vision recreates their controls with a camera.



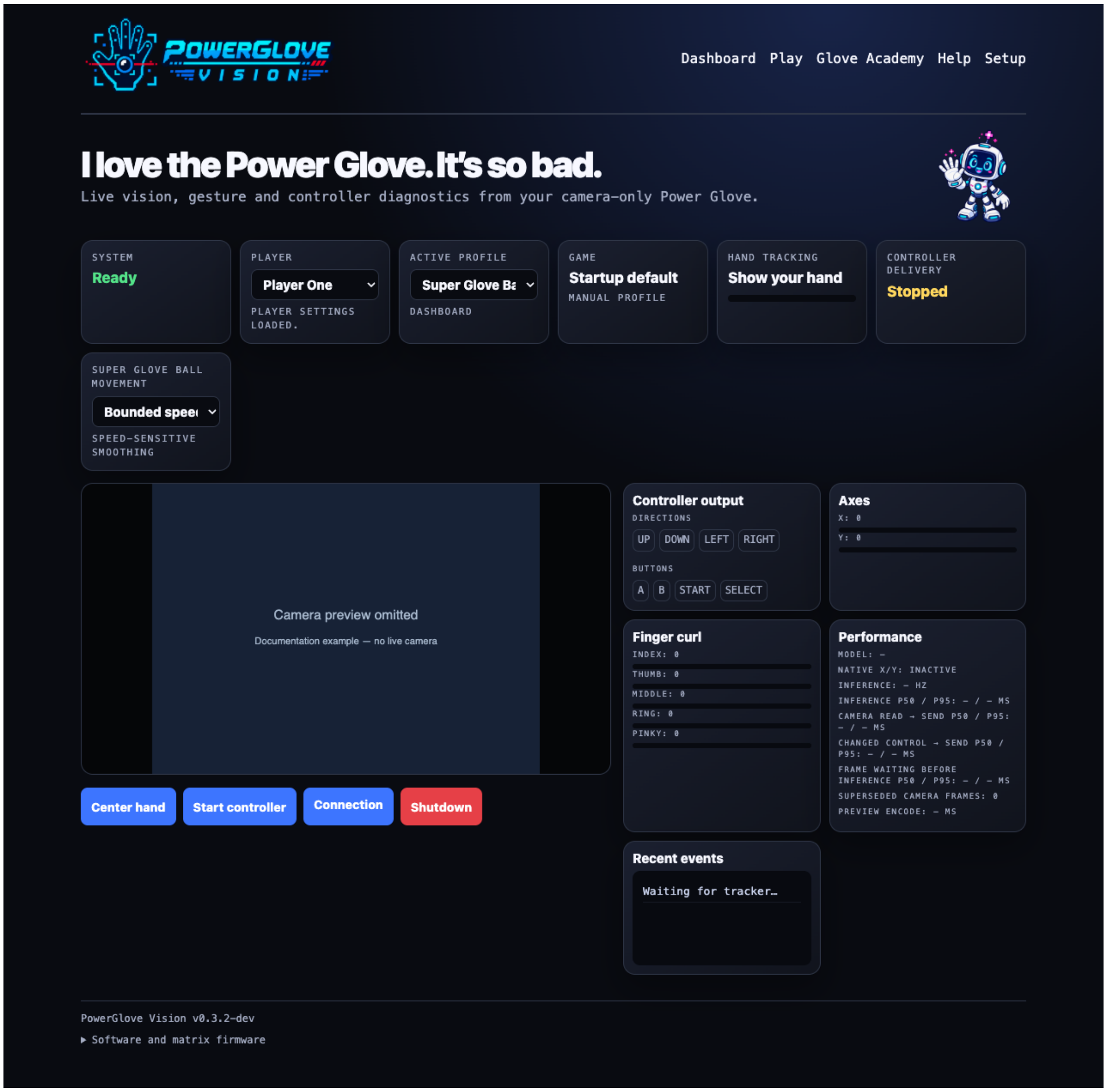
Pixel Pal, your arcade practice buddy

Pixel Pal says: Learn one gesture in Glove Academy, pick a program, then try a round. For a picture of each basic gesture, see [Gesture reference](#).

> **Pixel Pal's Extra-Digit Hunt:** Some glove illustrations have five fingers plus a thumb. Count every six-digit hand once per appearance, including repeated artwork. Pixel Pal reveals the answer at the back of this guide.

Choose a program and try it

1. Use the selector below to choose a program for your game or experiment.
2. Select that profile on Dashboard, wait for tracking, and check its gestures with controller delivery stopped.
3. Select **Start controller** when ready to play. Use the Gameplay Guide for game objectives and first-round exercises.
4. If the mapping suits another game, add its exact filename to the registry using Register games and select profiles.



Dashboard profile selector and controller diagnostics

To change sensitivity without changing a program's button assignments, open **Glove Academy -> Tune gestures**. Pixel Pal guides the adjustment; numerical thresholds are under **Advanced**. The matrix shows **T** during tuning and **L** during the sixteen learning lessons. Each player keeps saved progress toward **Glove Master**. See the illustrated [tuning walkthrough].

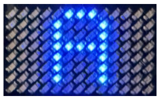
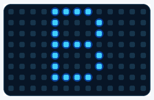
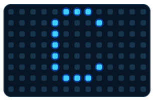
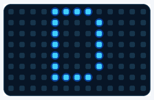
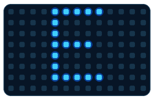
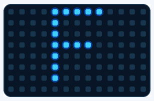
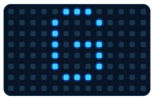
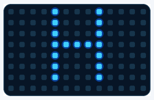
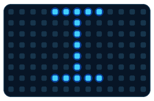
Where the programs came from

Bad Street Brawler contained nine configuration programs labelled A through I. The player loaded one into the Power Glove, switched off the NES, swapped to a different cartridge within roughly 30 seconds, and kept using the downloaded mapping.

These programs were not replacement firmware. They were small configurations for the glove's resident gesture interpreter. Each program mapped hand position, depth, wrist angle, and finger bends to ordinary NES controller inputs. The next game therefore did not need special Power Glove support.

PowerGlove Vision keeps all nine profiles ready at once. Select one on Dashboard or let RetroPie choose it when a game launches. You do not need to open Bad Street Brawler first.

Quick selector

Program	See it	Best fit	Main controls
A		Pinball	Two finger flippers, wrist tilt, combined-flipper mode
B		Joust	Steer by position; curl a finger to flap
C		Gyruss	Rotate by wrist angle; fire and bomb gestures
D		Challenge mode	Reversed directions with thumb/index buttons
E		Defender II	Ship movement, fire, smart bomb, evasive movement
F		Sesame Street 1-2-3	Open-hand Yes and closed-hand No
G		Gun Smoke	Walk by position; combine index curl and a forward push to fire
H		Training / general play	Familiar controls with pulsed buttons
I		Knight Rider / driving	Wrist steering, throttle, brake, and turbo

Menu gestures shared by every program

Gesture	See it	Controller result
Hold a V sign steadily		Start or pause after half a second
Briefly show a thumbs-up with the other fingers closed		Select

These menu poses suppress A/B attacks while they form. Some programs can still produce directional output from wrist, depth, or finger gestures. Keep your hand near its calibrated resting position, and return to a relaxed open hand after the command is recognized.

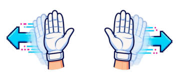
Program cards

A - Pinball rig

Gesture	See it	Controller result
Curl the index finger		Right flipper / A
Curl the thumb		Left flipper / Up
Roll the wrist left or right		Tilt / B
Pull the hand away from the camera		Toggle combined-flipper behaviour

Use this profile for pinball tables and games that benefit from two independent finger actions.

B - Joust rig

Gesture	See it	Controller result
Move the hand left or right		Steer left or right
Curl the thumb		B button; see the Joust play card for its in-game use
Curl the index or middle finger		Pulsed flap input

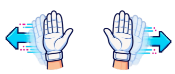
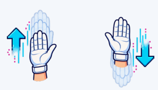
Use this profile for Joust and any game where rhythmic, repeated presses matter.

C - Gyruss rig

Gesture	See it	Controller result
Roll the wrist left or right		Rotate counter-clockwise or clockwise
Keep the index finger straight		Continuous fire
Pull the hand away from the camera		Launch a bomb

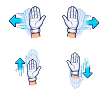
Use this profile for circular shooters and games with rotation plus rapid fire.

D - Mirror-world rig

Gesture	See it	Controller result
Move the hand left or right		Reversed right or left direction
Raise or lower the hand		Reversed down or up direction
Curl the thumb		First action button
Curl the index finger		Second action button

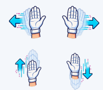
Use this profile for deliberate chaos, party challenges, or accessibility experiments that need inverted direction mappings.

E - Defender rig

Gesture	See it	Controller result
Move the whole hand		Move the ship
Curl the thumb		Fire
Roll the wrist left or right		Smart bomb
Curl the ring finger		Rapid horizontal movement

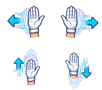
Use this profile for Defender II and multi-action shooters.

F - Yes / No rig

Gesture	See it	Controller result
Close every finger into a fist		No
Move an open hand in any direction		Yes

Use this profile for Sesame Street 1-2-3 and simple choice-driven games.

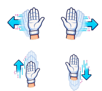
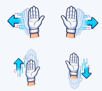
G - Gun Smoke rig

Gesture	See it	Controller result
Move the whole hand		Walk using horizontal and vertical hand movement
Curl the index finger		Fire
Roll the wrist		Add left or right movement
Curl the thumb and ring finger		Suppress all ordinary controller output

A forward push sends B. Combine it with index curl (A) to send A+B.

Use this profile for Gun Smoke and shooters with movement plus directional fire.

H - Training rig

Gesture	See it	Controller result
Move the whole hand		Conventional directions
Curl the thumb		Pulsed B
Curl the index finger		Pulsed A
Return the hand to center		Release directional input

Use this profile for learning the system or giving an unmapped game a sensible general-purpose starting point.

I - Driving rig

Gesture	See it	Controller result
Roll the wrist left or right		Steer
Curl the index finger		Throttle
Lower the hand		Brake
Push the hand toward the camera		Turbo
Curl the thumb		Auxiliary action

Use this profile for Knight Rider and driving games that need steering, speed, and one extra action.

How PowerGlove Vision selects a program

Choose the current profile on Dashboard; choose the saved startup profile on Setup. Automatic selection matches the complete ROM filename, including its extension but excluding its folder path, against `/etc/powerglove/games.json` on RetroPie. Matching ignores letter case.

The launch hook sends an authenticated profile request. The PowerGlove Vision Controller releases held controls, changes the mapping, reuses the saved calibration, and acknowledges the new profile on its blue matrix. If no valid calibration is saved, it collects an initial reference while you hold your open hand still in a comfortable resting position. It uses 24 geometrically valid observations; repeating the same center, distance, and wrist pose produces a similar rather than bit-for-bit identical reference.

```
JSON
{
  "games": {
    "Joust (USA).zip": "program_b",
    "Gyruss (USA).nes": "program_c",
    "Gun.Smoke (USA).7z": "program_g"
  }
}
```

Merge entries into the existing `games` object; do not replace other registered games. The matrix displays `A` through `I`. When an unknown game starts, the launch hook turns gesture control off so the previous game's mapping does not remain active.

Practise safely

1. Open `http://UNO-Q-NAME.local:8088/learn`.
2. Keep your whole hand visible. Select **Center hand** on first use or when your resting position produces unwanted movement, then hold still.
3. Move slowly until the intended gesture is recognized consistently.
4. Open Dashboard to compare hand motion with generated controller output.
5. Start controller delivery only when ready to play.

Glove Academy mode works without RetroPie and automatically starts the camera while suppressing controller output. This also works while **Gestures off** is the selected profile. Leaving Glove Academy restores the selected mode, so it is a safe place to build muscle memory without changing the saved or active selection.

What remains special

The nine profiles emit useful conventional controller inputs, so ordinary NES cores can use them today. Bad Street Brawler's Glove Zap also works through standard inputs: its dedicated profile emits a short simultaneous Left + Right pulse. FCEUmm must allow opposing directions for that game. Richer native glove integration remains separate work; PowerGlove Vision preserves analogue and finger data for it and does not modify ROM images.

For installation, secure pairing, RetroArch mapping, and troubleshooting, see the Installation Guide.

Project note

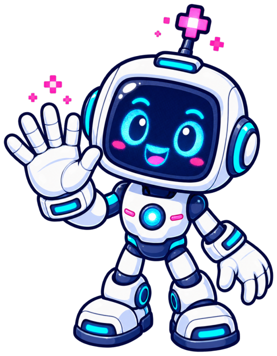
PowerGlove Vision is an independent MIT-licensed hobbyist project by Iain Bennett. Nintendo, NES, Power Glove, Bad Street Brawler, and other marks belong to their respective owners. No ROM images are distributed with this project.

Personal sensitivity and practice

Glove Academy has sixteen lessons, including **Glove Zap**, **Pull Back**, both wrist rolls, close hand, and menu guard. In **Tune gestures**, tell Pixel Pal whether a hand is new, a gesture is difficult, a gesture happens accidentally, or movement feels off-centre. The wizard waits for one second of clear, stable tracking before the user starts its countdown. Pose, direction, and roll steps take two seconds; Glove Zap and Pull Back use three short motion-and-return repetitions so a still near or far hand cannot teach the motion. A preview must pass two uses and releases plus three neutral seconds before it can be saved.

Saved thresholds apply across profiles, including these programs. Movement controls use separate activation and release thresholds, while program-specific button pulses and mappings remain in effect. Advanced values and selective reset remain available without exposing them in the normal family workflow. Practice, personalization, preview, and diagnostics pause controller delivery.

Pixel Pal's Extra-Digit Hunt answer



Pixel Pal reveals the Extra-Digit Hunt answer

Pixel Pal's answer: 3 six-digit hands.

They appear once each in Program B's finger-curl illustration, Program E's finger-curl illustration, and Program F's closed-hand illustration. Every appearance counts, even when the same artwork returns.